

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 21

18-07-2026

21. WRITE A PROLOG PROGRAM TO IMPLEMENT TOWERS OF HANOI

AIM

To write and execute a Prolog program to implement the Tower of Hanoi problem using recursion. The objective is to move all disks from a source peg to a destination peg using an auxiliary peg while following the rules of the Tower of Hanoi.

ALGORITHM

1. Start the program.
2. Consider three pegs: source, auxiliary, and destination.
3. If there is only one disk, move it directly from source to destination.
4. For N disks, first move N-1 disks from source to auxiliary.
5. Move the remaining largest disk from source to destination.
6. Move the N-1 disks from auxiliary to destination.
7. Repeat the process recursively until all disks are moved.
8. Display every disk movement.
9. Stop the program.

PROLOG PROGRAM

% Write a Prolog program to implement Towers of Hanoi.

hanoi(1, Source, Destination, _) :-

write('Move disk from '),

write(Source),

```
write(' to '),  
write(Destination),  
nl.
```

hanoi(N, Source, Destination, Auxiliary) :-

```
    N > 1,  
    N1 is N - 1,  
    hanoi(N1, Source, Auxiliary, Destination),  
    write('Move disk from '),  
    write(Source),  
    write(' to '),  
    write(Destination),  
    nl,  
    hanoi(N1, Auxiliary, Destination, Source).
```

SAMPLE INPUT

?- hanoi(3, left, right, middle).

SAMPLE OUTPUT

```
Move disk from left to right  
Move disk from left to middle  
Move disk from right to middle  
Move disk from left to right  
Move disk from middle to left  
Move disk from middle to right  
Move disk from left to right
```

true.

RESULT

Thus, the Tower of Hanoi problem was successfully implemented using recursion in Prolog. For three disks, the program generated the required seven movements and successfully transferred all disks from the source peg to the destination peg.